

18. Write each of the following three statements in symbolic form and determine which pairs are logically equivalent. Include truth tables and a few words of explanation.

If it walks like a duck and it talks like a duck, then it is a duck.

Either it does not walk like a duck or it does not talk like a duck, or it is a duck.

If it does not walk like a duck and it does not talk like a duck, then it is not a duck.

First, we assign plausible statements to variables, which can be either true or false.

p = It walks like a duck

q = It talks like a duck.

r = It is a duck.

Followed by translating the statements into symbolic form, as represented below:

Statement	Symbolic form
“If it walks like a duck and it talks like a duck, then it is a duck”	$(p \wedge q) \rightarrow r$
“Either it does not walk like a duck or it does not talk like a duck, or it is a duck.”	$\sim(p \wedge q) \vee r$
“If it does not walk like a duck and it does not talk like a duck, then it is not a duck.”	$\sim(p \vee q) \rightarrow \sim r$

Let's create a truth table for $(p \wedge q) \rightarrow r$

p	q	$p \wedge q$	r	$(p \wedge q) \rightarrow r$
T	T	T	T	T
T	T	T	F	F
T	F	F	T	T
T	F	F	F	T
F	T	F	T	T
F	T	F	F	T
F	F	F	T	T
F	F	F	F	T

Let's look at the definition of the conditional statement (according to Epp):

Definition

If p and q are statement variables, the **conditional** of q by p is “If p then q ” or “ p implies q ” and is denoted $p \rightarrow q$. It is false when p is true and q is false; otherwise it is true. We call p the **hypothesis** (or **antecedent**) of the conditional and q the **conclusion** (or **consequent**).

By this definition, we can conclude that the truth set for $(p \wedge q) \rightarrow r$ is false when the hypothesis $(p \wedge q)$ is true and the conclusion r is false; otherwise, it is true.

Let's also create a truth table for $\sim(p \wedge q) \vee r$

p	q	$\sim(p \wedge q)$	r	$\sim(p \wedge q) \vee r$
T	T	F	T	T
T	T	F	F	F
T	F	T	T	T
T	F	T	F	T
F	T	T	T	T
F	T	T	F	T
F	F	T	T	T
F	F	T	F	T

And finally, another truth table for $\sim(p \vee q) \rightarrow \sim r$

p	q	$\sim(p \vee q)$	$\sim r$	$\sim(p \vee q) \rightarrow \sim r$
T	T	F	F	T
T	T	F	T	T
T	F	F	F	T
T	F	F	T	T
F	T	F	F	T
F	T	F	T	T
F	F	T	F	F
F	F	T	T	T

Now we can compare two three statement forms in a truth table:

$(p \wedge q) \rightarrow r$	$\sim(p \wedge q) \vee r$	$\sim(p \vee q) \rightarrow \sim r$
T	T	T
F	F	T
T	T	T
T	T	T
T	T	T
T	T	T
T	T	F
T	T	T

By comparing the truth sets from each statement form, we can deduce that there're two logically equivalent pairs: $(p \wedge q) \rightarrow r \equiv \sim(p \wedge q) \vee r$, whereas $\sim(p \vee q) \rightarrow \sim r$ is not logically equivalent to the aforementioned.

Note: What is especially interesting, is that the truth set for $\sim(p \vee q) \rightarrow \sim r$ is reversed, comparatively to $(p \wedge q) \rightarrow r \equiv \sim(p \wedge q) \vee r$

Solution: $(p \wedge q) \rightarrow r$ and $\sim(p \wedge q) \vee r$ are logically equivalent: $(p \wedge q) \rightarrow r \equiv \sim(p \wedge q) \vee r$